

PRADAAP SHIVA KUMAR SHOBHA

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer delivering high-impact **distributed systems**: **\$2B+** transaction platform, **\$500K cost reduction**, 99.8% order completion rate, and **85% reduction in manual operations**. Proven expertise in **Azure microservices (AKS, Service Bus)**, event-driven architecture (Kafka), **Node.js/Java backends**, and zero-downtime deployments. Specialized in modernizing legacy systems to cloud-native architectures.

PROFESSIONAL EXPERIENCE

Full Stack Software Engineer

Hub Group Inc | Jan 2024 - Present

- Led design and implementation (**with 1 senior engineer and 2 SDEs**) of **microservices**-based billing platform processing **\$2B+ annual invoice volume**, automating invoice processing and **ERP integration** while retiring legacy **COBOL** systems.
- Contributed to architecture and implementation of migration (**over 10 months, across 12 microservices**) from **monolithic COBOL** to containerized **Node.js microservices** on **Azure Kubernetes Service (AKS)**, reducing estimated annual costs by **\$500K** while enabling independent scaling and **zero-downtime deployments**.
- Engineered **REST API** integration between Order Management platform and TMS using custom **ETL pipelines** processing **50K+ daily orders**, eliminating manual data transfers and driving an **estimated \$1M revenue growth** through operational efficiency.
- Designed and built real-time order failure detection system using **Azure Service Bus topics** and automated recovery workflows, improving order completion rate to 99.8% and reducing manual intervention by **85%** across **100K+ monthly transactions**.
- Redesigned **distributed API architecture** to support **10x traffic growth**, implementing **Redis clustering** and multi-tier **caching strategies** to handle **500K+ monthly requests**, reducing P95 response time by **60% (800ms → 320ms)**.
- Built **anomaly detection system** using threshold-based alerting and ML-augmented baselines (TensorFlow), integrated with Elasticsearch/Kibana dashboards to **detect 15+ critical incidents**, reducing **MTTD** to under 2 minutes.
- Automated **zero-downtime deployment** pipeline using **Kubernetes rolling updates via Jenkins**, coordinating releases across 8 microservices with automated health checks, and rollback mechanisms, **reducing release cycle (4 hours → 1 hour)**.

Software Engineer

Cerner Corporation [Oracle Health] | Apr 2021 - Aug 2022

- Implemented **HL7 v2.x** and **FHIR** message handlers processing **150K+ daily patient transactions**, reducing interface transmission errors by **22%** through enhanced validation logic and error handling across production **EHR integrations**.
- Created custom **Discern Rules** and **SQL**-based reporting solutions against **Oracle Database** for revenue cycle and scheduling workflows, delivering **15 production reports** serving **500+ clinical and operational staff** members daily.
- Designed an automated clinical reporting framework using **CCL** and **Oracle Database**, deployed across **8 hospital clients** serving **10K+ clinicians**, reducing manual data reconciliation by **45%** (saving **20 staff-hours/week** per site).

Programmer Analyst

Cognizant | Sep 2020 - Apr 2021

- Built full-stack financial platform using **React.js** and **Spring Boot microservices** with **OAuth2/JWT** authentication and **role-based access control (RBAC)**, serving **10K+ daily active users** while maintaining **SOC2 compliance** and **99.9% uptime**.
- Engineered high-throughput **RESTful APIs** using **Spring Security** with **Redis caching** and custom interceptors, reducing API latency by **35%** and implementing **rate limiting** for critical World Bank integration endpoints processing **50K+ daily requests**.
- Architected hybrid database solution using **MySQL sharding strategies** and **indexing**, improving query performance by **40%** while ensuring ACID compliances, supporting **100M+ records** across distributed financial transaction systems.

TECHNICAL SKILLS

Programming Languages: JavaScript, Python, TypeScript, Java, SQL, CCL, C++

Frontend Technologies: React.js, Vue.js, Next.js, Angular, HTML5, CSS3, State Management (Redux, Vuex)

Backend & APIs: Node.js, Express, Spring Boot, REST APIs, GraphQL, API Design, WebSockets

System Design & Architecture: Distributed Systems, Microservices Architecture, Event-Driven Architecture, Service-Oriented Architecture (SOA), System Scalability, High Availability, Load Balancing, Service Mesh, API Gateway, Fault Tolerance

Cloud & DevOps: Azure (Azure Kubernetes Service (AKS), Service Bus, Blob Storage, DevOps, Logic Apps), AWS (EC2, Lambda, S3, RDS, DynamoDB, API Gateway, CloudWatch), Kubernetes, Docker, CI/CD Pipelines, Jenkins, GitHub Actions, Infrastructure as Code (Terraform), Container Orchestration

Databases & Data Storage: MongoDB, PostgreSQL, MySQL, Oracle Database, SQL Server, Redis (Caching)

Message Queues & Streaming: Azure Service Bus, Apache Kafka, RabbitMQ, Event Streaming, Asynchronous Processing

Security & Authentication: OAuth2, JWT, Role-Based Access Control, API Security, Data Encryption, HIPAA/SOC2 Compliance

Observability & Reliability: Elasticsearch, Kibana, Prometheus, Logging (ELK Stack), Metrics & Monitoring, Alerting, Distributed Tracing, Performance Optimization, Application Reliability Engineering, Incident Response, On-Call Rotation

Testing & Quality: Unit Testing, Integration Testing, Test-Driven Development (TDD), SonarQube, Automated Testing

AI & Machine Learning: FastAPI, TensorFlow, LLM Workflows, GenAI Application Integration, Agent Orchestration

Tools & Practices: Git, Agile/Scrum, RESTful Design Patterns, Code Review, Technical Documentation, System Troubleshooting

EDUCATION

Master of Computer Science - Illinois Institute of Technology, USA